



Experimental and modeling studies of quadricyclane and 2-ethylnorbornane pyrolysis from atmospheric to high pressure

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ABSTRACT

To improve the understanding of cyclopentadiene-based high-density liquid hydrocarbon fuels combustion chemistry, the pyrolysis experiments of 1,3-cyclopentadiene ($c\text{-C}_5\text{H}_6$), quadricyclane (QC) and ethylnorbornane (EthNB, reported in our recent work: Wang et al., 2020), were performed in a flow reactor at atmospheric and high pressure over 348–1173 K. The mole fraction profiles of products in three fuels pyrolysis were obtained using online GC-MS/FID. Based on theoretical calculations and literature research, a universal kinetic model (397 species and 1522 reactions) of QC and EthNB, incorporating the sub-mechanisms of $c\text{-C}_5\text{H}_6$ and polycyclic aromatics, was constructed in this work. It was validated against the present data and pyrolysis data of $c\text{-C}_5\text{H}_6$ as well as norbornane in literature with reasonable reproducibility. The rate of production analysis shows that the retro-Diels-Alder reaction to $c\text{-C}_5\text{H}_6$ plus C_2H_2 is always the dominant path to 2,5-norbornadiene (NBD, the only isomer product of QC) consumption at atmospheric and high pressures, whereas the contribution of isomerization to 1,3,5-cycloheptatriene increases with the increasing pressure and is almost equal to that of the former at high pressure and low conversion. The aromatics formation channels in QC pyrolysis are affected by the presence of C_2H_2 and different from those of $c\text{-C}_5\text{H}_6$ pyrolysis, especially under high pressure. For EthNB decomposition, the open-ring and H-abstraction reactions play a dominant role. The C2-C6 alkene products are formed via the decomposition of ethylnorbornyl radicals and the further reactions of these alkenes are the precursors of aromatics in EthNB pyrolysis. Although QC and EthNB have a similar U-shaped carbon skeleton structure, the difference is that the former has an extra four-membered ring with more strain energy, which leads a lower initial decomposition temperature. Similarly, the extra $\text{C}=\text{C}$ double bonds, in NBD compared to EthNB, result in more $c\text{-C}_5\text{H}_6$ and C_2H_2 formation and further increase the growing tendency of initial PAH.

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1. Introduction

As one of the most important high-density liquid hydrocarbon fuels (HDLHFs), cyclopentadiene (CPD, $c\text{-C}_5\text{H}_6$) based HDLHFs have attracted great attentions from combustion community due to their great potential as engine fuels or fuel additives [1–5]. They are usually synthesized from the Diels-Alder addition reactions of CPD with olefin monomers, followed by hydrogenation and isomerization reactions. For example, the reactions of CPD with acetylene, butadiene, cyclopentadiene and dicyclopentadiene can ultimately form four CPD-based HDLHFs, namely quadricyclane (QC, C_7H_8 , 0.98 g/cm^3), ethylnorbornane (EthNB, C_9H_{16} , 0.86 g/cm^3), exo-tetrahydrodicyclopentadiene (exo-THDCPD, $\text{C}_{10}\text{H}_{16}$, 0.94 g/cm^3)

and tetrahydrotricyclopentadiene (THTCPD, $\text{C}_{15}\text{H}_{22}$, 1.02 g/cm^3) [6–10]. Besides high density, high volumetric net heat of combustion and good cryogenic characteristics including freezing point and viscosity of these fuels are also critical to meet the needs of aerospace vehicles in the low temperature environment [11,12]. Moreover, CPD is also known as a key intermediate in the formation of polycyclic aromatic hydrocarbons (PAHs) [13]. Consequently, experimental and kinetic modeling research on combustion and pyrolysis chemistry of CPD-based HDLHFs are of great significance for the practical application of fuels.

Among CPD-based HDLHFs, considering the extensive studies of exo-THDCPD and the extremely high viscosity limit of THTCPD (510 mPa·s, 233 K [14]), QC (3.06 mPa·s, 233 K [14]) and EthNB (5.47 mPa·s, 233 K [4]) are chosen in this work to investigate the pyrolysis behaviors and obtain insight into the effects of fuel structure on its reactivity and PAH formation, independent of fuel

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